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ORIGINAL RESEARCH

National treatment practices in the management of infectious intracranial aneurysms and infective endocarditis

Amit Singla,[1] Kyle Fargen,[1] Spiros Blackburn,[1] Dan Neal,[1] Tomas D. Martin,[2] Phillip J Hess,[2] Thomas M Beaver,[2] Charles T Klodell,[2] Brian Hoh[1]

[1]Department of Neurosurgery, University of Florida, Gainesville, Florida, USA
[2]Department of Thoracic and Cardiovascular Surgery, University of Florida, Gainesville, Florida, USA

Correspondence to
Dr Brian Hoh, Department of Neurosurgery, University of Florida, PO Box 100265, Gainesville, FL 32610, USA; brian.hoh@neurosurgery.ufl.edu

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ABSTRACT

Introduction There is an absence of widely accepted guidelines for the management of infectious intracranial aneurysms (IIAs) owing to a dearth of high-quality evidence in the literature.

Objective To better define the incidence of IIAs, treatment practices, and patient outcomes by performing a Nationwide Inpatient Sample (NIS) database query.

Methods We queried the NIS database from 2002 to 2011 for all patients with the *primary diagnosis* of infectious endocarditis (IE), subarachnoid hemorrhage (SAH), or unruptured cerebral aneurysm by ICD-9-CM codes. ICD-9 procedure codes were used to identify patients undergoing neurosurgical or cardiothoracic procedures.

Results The query identified 393 patients with *primary diagnosis* of IE, SAH or unruptured cerebral aneurysm treated during 2002–2011. The mean age of the patients was 53.5 years; 244 (62%) were male. The majority of patients presented with SAH (361; 91.9%). Only 73 (18.6%) patients underwent neurosurgical coiling or clipping for IIA. Of patients undergoing a neurosurgical procedure, 65 had SAH (constituting only 18% of patients with SAH) and 8 had unruptured aneurysms (constituting only 25% patients with unruptured aneurysms). Cardiac procedures were performed in only 72/393 patients (18.3%) patients. Only 67 (18.6%) of the patients with SAH and 5 (15.6%) with unruptured aneurysms underwent a cardiac corrective surgical procedure. Mortality was significantly higher in those patients managed conservatively (26.7%) than in those who underwent clipping or embolization (15.1%; $p<0.001$).

Conclusions In this NIS database study, the majority of patients with IIAs were managed non-operatively, regardless of rupture status. Further investigation is warranted to standardize the management of these lesions.

INTRODUCTION

Intracranial infectious aneurysms (IIAs), previously referred to as 'mycotic' aneurysms, are a relatively rare entity comprising approximately 0.7–5.4% of all cerebral aneurysms.[1 2] IIAs are reported to occur in 1–10% of patients with infective endocarditis (IE) and about 65% of patients with IIAs have coexistent IE.[3 4] Most IIAs are thought to occur via hematogenous spread of septic emboli from valvular vegetations in the setting of IE; other less common hematogenous causes may include a history of intravenous drug use and nosocomially acquired blood stream infections. They may also occur from direct extravascular invasion from a nearby intracranial infectious source such as meningitis, cavernous sinus thrombophlebitis, or orbital cellulitis.[3 5–7]

The coexistence of IE in the majority of patients with IIA makes their treatment more challenging with respect to timing, the need for anticoagulation and the concern about heart failure. In general, the treatment is dependent on the rupture status of the IIA. Antibiotic therapy for the causative pathogen is traditionally considered the standard treatment for *unruptured* IIAs, whereas surgical or endovascular intervention may be considered for *ruptured* aneurysms. Given the high rupture rates of IIAs and the high morbidity and mortality associated with rupture, timely detection and treatment seems prudent.[8] Treatment strategies may include antibiotics alone, surgery or endovascular treatment of the IIA, cardiac surgery for the infected cardiac valve, or some combination of these treatments.

The optimal management strategy for IIAs is not well understood owing to their complexity and rarity. Furthermore, information about the efficacy and safety of treatment strategies is often limited to small case series. Because IIAs are so rare, there are insufficient patient series from single centers to define the optimal treatment for these challenging patients—in particular, the question of whether patients should undergo IIA treatment, cardiac valve replacement surgery, or both, and the timing and which should be performed first. To better define the incidence of IIAs, national neurosurgical and cardiothoracic treatment practices for IIAs, and patient outcomes associated with these lesions, we performed a Nationwide Inpatient Sample (NIS) database query of patients treated for IIA in the USA over a 10-year period.

METHODS

We queried the NIS database from 2002 to 2011 for all patients with the *primary diagnosis* of IE, subarachnoid hemorrhage (SAH) or unruptured cerebral aneurysm. Using the combined International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) codes for diagnoses of endocarditis (391.1, 421.0, 421.1, 421.9, 424.9, 424.91, or 424.99) and SAH (430) or unruptured cerebral aneurysm (437.3), the patients with IIAs were identified. Then using the ICD-9 procedure

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codes for clipping (39.51) and coiling (39.52, 39.72, 39.75, 39.76, or 39.79) of cerebral aneurysms, the patients who received neurosurgical intervention were computed. Similarly, ICD-9 codes for cardiac procedures (35.10, 35.11, 35.12, 35.13, 35.14, 35.20, 35.21, 35.22, 35.23, 35.24, 35.25, 35.26, 35.27, 35.28, or 35.72) were used to identify the patients who received cardiac procedures during their hospitalization.

The outcomes were defined based on a previously published[9] dichotomous categorization as good or poor as described below:

- good=discharged home, transferred to short-term hospital, discharged home under home healthcare, left against medical advice, discharged to home IV provider, discharged to inpatient rehabilitation facility, discharged to another institution for outpatient care.
- poor=discharged to skilled nursing facility, discharged to intermediate- care facility, died in hospital, discharged to home hospice, discharged to hospice facility, discharged to hospital-based Medicare-approved swing bed, discharged to Medicare certified long-term care hospital, discharged to Medicaid-certified nursing facility.

The All Patient Refined-Diagnosis Related Group (APR-DRG) mortality risk algorithm was used as a surrogate marker of a patient's clinical condition to assess its effect on the treatment decision. The APR-DRG system is a joint development of 3M Health Information Systems (Salt Lake City, Utah, USA) and the National Association of Children's Hospitals and Related Institutions.[10] The APR-DRG scores are calculated from discharge billing codes and are based on primary and secondary discharge diagnosis, age, and pre-existing medical conditions.[11 12] The severity of illness subclasses based on the risk of mortality are minor, moderate, major, and extreme severity of illness. This algorithm is known to be a reliable and valid risk-of-mortality adjustment system.[13]

Statistical analysis

An SAS software package (Cary, North Carolina, USA, V.9.3) was used to calculate all descriptive statistics and conduct all statistical tests. For the analysis, χ^2 or Fisher's exact tests were used to compare categorical variables in the clipping, coiling, and no treatment groups, and analysis of variance or the Kruskal–Wallis test to compare continuous variables in these groups. A linear regression model was used for analysis of procedure volume by year.

Multivariable analysis

Generalized estimating equations logistic regression was performed to assess which variables were predictive of receiving a brain procedure when controlling for other factors. Brain procedure (yes or no) was taken as the outcome, and APR-DRG severity score, aneurysm type (ruptured or unruptured), age, gender, income status, opioid use, cardiac procedure, hospital size (small/medium or large), hospital teaching status, and total number of comorbidities were taken as fixed factors. For multivariate analysis of cardiac procedure, the cardiac procedure was taken as the outcome and brain procedure was taken as a covariate.

RESULTS

The query identified 393 patients with a *primary diagnosis* of IIA and IE treated during 2002–2011. The mean age of the patients was 53.5 years; 244 (62%) were male. The majority of patients (361/393; 91.9%) presented with SAH from aneurysm rupture (table 1). Unruptured IIAs were detected in 32 (8.1%) patients. Only six (1.5%) patients (five with SAH and one with unruptured IIA) underwent both aneurysm coiling or clipping and a cardiac procedure during the same admission.

Treatment practices: neurosurgical

Seventy-three of 393 (18.6%) patients underwent neurosurgical embolization or clipping for IIA. In the patient group with SAH, an intervention was performed in 65/361 (18%). In the unruptured aneurysm patient group, an intervention was performed in 8/32 (25%). Of patients undergoing a neurosurgical procedure, 65/73 (89%) patients had SAH and 8/73 (11%) patients had unruptured aneurysms. Twenty-seven of 73 (37%) patients underwent surgical clipping and 46/73 (63%) patients underwent endovascular coiling for IIAs. Factors associated with an increased likelihood of neurosurgical treatment included younger age, higher income status, and larger hospital size. The likelihood of neurosurgical intervention significantly decreased with age (OR multiplies by 0.97 for each additional year, 95% CI (0.950 to 0.983), p<0.0001) and significantly increased with income status (OR multiplies by 1.3 for each step up in income level, 95% CI (1.02 to 1.67), p=0.037). In addition, patients treated at large hospitals were significantly more likely to receive neurosurgical procedures than those in smaller hospitals (OR multiplies by 2.4 for each step up in bed size (small to medium to large), 95% CI (1.23 to 4.84), p=0.010). Absence of opioid use (p=0.029) and cardiac procedure (p=0.011) were associated with an increased likelihood of undergoing neurosurgical treatment. APR-DRG severity score, aneurysm type (ruptured or unruptured), gender, hospital teaching status were not associated with increased likelihood of neurosurgical treatment (table 1).

Treatment practices: cardiothoracic surgery

Cardiac procedures were performed in only 72 of the 393 patients (18.3%) patients. Only 67 (18.6%) of the patients with SAH and 5 (15.6%) with unruptured aneurysms underwent a cardiac corrective surgical procedure. Most of the patients (92.0%) who received a cardiac procedure did not undergo a neurosurgical procedure during that hospitalization. Several factors were associated with an increased likelihood of undergoing a cardiac procedure during the hospitalization. The likelihood of undergoing a cardiac procedure significantly decreased with age (OR multiplies by 0.97 for each additional year, 95% CI (0.955 to 0.986), p=0.0002). Further, the likelihood of a cardiac procedure significantly increased with income status (OR multiplies by 1.4 for each step up in ZIP code income, 95% CI (1.13 to 1.80), p=0.003). In addition, patients treated in non-teaching hospitals were significantly less likely to have a cardiac procedure (OR=0.24, CI (0.087 to 0.648), p=0.0017). The likelihood of a cardiac procedure significantly increased with hospital size (OR multiplies by 1.6 for each step up in bed size (small to medium to large), 95% CI (0.943 to 2.87), p=0.0795). The likelihood of a cardiac procedure significantly increased with APR-DRG Severity score (OR multiplies by 4.4 for each unit increase in score, 95% CI (1.70 to 11.3), p=0.002). Patients not receiving a brain procedure were significantly more likely to have a cardiac procedure (OR=4.5, 95% CI (1.62 to 12.8), p=0.004). Opioid use, aneurysm type (ruptured or unruptured), gender were not associated with an increased likelihood of neurosurgical treatment (table 2).

Treatment practices: temporal trends

The proportion of patients receiving neurosurgical or cardiothoracic procedures has remained roughly constant over the years 2002–2011 even though the total number of patients who

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Table 1 Demographic and outcome data for patients with infectious intracranial aneurysms based on neurosurgical treatment

	All patients N (%) or mean (SD)	No treatment N (%) or mean (SD)	Clipping or embolization N (%) or mean (SD)	p Value
Patient and hospital factors (row percentages)				
Patients (n)	393	320	73	N/A
Aneurysm type				
SAH	361 (91.9)	296 (82.0)	65 (18.0)	0.34
Unruptured	32 (8.1)	24 (75.0)	8 (25.0)	
Age				
Mean (SD)	53.5 (17.7)	54.7 (17.5)	48.3 (17.9)	0.02
Gender, n (%)				
Male	244 (62.1)	204 (83.6)	40 (16.4)	0.18
Female	149 (37.9)	116 (77.9)	33 (22.1)	
Median income in ZIP code				0.28
All patient redefined DRG: severity of illness subclass				
Moderate loss of function	1 (0.3)	1 (100)	0 (0)	0.80
Major loss of function	89 (22.8)	73 (82.0)	16 (18.0)	
Extreme loss of function	301 (77.0)	244 (81.1)	57 (18.9)	
Opioid use				
No	364 (92.6)	293 (80.5)	71 (19.5)	0.13
Yes	29 (7.4)	27 (93.1)	2 (6.9)	
Any cardiac procedure				
No	321 (81.7)	254 (79.1)	67 (20.9)	0.012
Yes	72 (18.3)	66 (91.7)	6 (8.3)	
Total comorbidities				
Mean (SD)	3.1 (2.0)	3.1 (2.0)	2.9 (2.0)	0.31
Hospital bed size				
Small	27 (6.9)	24 (88.9)	3 (11.1)	0.005
Medium	68 (17.3)	63 (92.6)	5 (7.4)	
Large	297 (75.8)	233 (78.5)	64 (21.5)	
Hospital teaching status				0.19
Outcomes (column percentages)				
Mortality	97 (24.7)	86 (26.9)	11 (15.1)	0.036
Poor outcome (54 observations missing)	166 (49.0)	142 (49.3)	24 (47.1)	0.88
Length of stay Mean (SD)	18.8 (18.1)	17.6 (16.6)	24.3 (22.8)	0.002
Total charges*				
(Mean (SD)	199.7 (214.7)	171.0 (177.8)	324.2 (301.4.2)	<0.0001

*adjusted 3%/year for inflation in $1000.
DRG, Diagnosis Related Group; SAH, subarachnoid hemorrhage.

underwent the procedures has increased, largely owing to an increasing number of cases of endocarditis with IIA. Linear regression models gave the following results:

- The number of brain procedures performed increased significantly with time (p=0.004). The number of procedures in the NIS has been increasing on average by 0.65 per year (95% CI (0.265 to 1.04)). The proportion of patients receiving brain procedures is not associated with year (p=0.145) (figure 1).
- The number of cardiac procedures performed was not associated with year (p=0.27). The proportion of patients receiving a cardiac procedure is not associated with year (p=0.84) (figure 2).

Mortality

Overall mortality was 24.7% in all patients included in this analysis. Mortality was 97/361 (36.7%) in patients with SAH and 0/32 (0%) in those with unruptured aneurysms (p=0.0001). Mortality was significantly higher in those patients managed conservatively (26.7%) than in those undergoing clipping or coiling (15.1%); p<0.001. Controlling for APR-DRG severity, the OR of mortality for conservative treatment versus neurosurgical intervention for IIAs=2.6, CI (1.40 to 4.91), p=0.0026. The small number of patient deaths in the treatment group precluded multivariate analysis for other factors. Mortality was 19.6% (9/46) in patients who underwent coiling and 7.4% (2/27) patients who underwent surgical clipping (p=0.19). The small number of patient deaths in the groups precluded multivariate analysis.

Clinical outcomes

Controlling for APR-DRG severity, neurosurgical intervention/conservative treatment is not associated with poor outcome (p=0.3325). Poor outcomes were reported in 7 patients in the clipping group and 17 patients in coiling group (p=1). Controlling for APR-DRG severity and comorbidities, there was no significant difference in the poor outcomes between the coiling and clipping groups.

Length of stay and hospital charges

Length of stay and hospital charges were significantly higher in patients who underwent cardiac or neurosurgical interventions

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Table 2 Demographic and outcome data for patients with infectious intracranial aneurysms based on cardiac treatment

	All patients N (%) or mean (SD)	No cardiac procedureN (%) or mean (SD)	Cardiac procedure N (%) or mean (SD)	p Value
Patient and hospital factors (row percentages)				
Patients, n (%)	393	321 (81.7)	72 (18.3)	N/A
Aneurysm type				
SAH	361 (91.9)	294 (81.4)	67 (18.6)	0.815
Unruptured	32 (8.1)	27 (84.4)	5 (15.6)	
Age				
Mean (SD)	53.5 (17.7)	54.6 (17.9)	48.3 (15.6)	0.006
Median (IQR) (range)	54 (41–67) (0–92)	55 (42– 69) (0–92)	49 (34.5–61.5) (22–82)	
Gender, n (%)				
Male	244 (62.1)	194 (79.5)	50 (20.5)	0.180
Female	149 (37.9)	127 (85.2)	22 (14.8)	
Median income in ZIP code				0.142
All patient redefined DRG: severity of illness subclass				
Moderate loss of function	1 (0.3)	1 (100)	0 (0)	
Major loss of function	89 (22.8)	82 (92.1)	7 (7.9)	
Extreme loss of function	301 (77.0)	237 (78.7)	64 (21.3)	0.006
Opioid use				
No	364 (92.6)	299 (82.1)	65 (17.9)	0.453
Yes	29 (7.4)	22 (75.9)	7 (24.1)	
Any brain procedure				
No	320 (81.4)	254 (79.4)	66 (20.6)	0.012
Yes	73 (18.6)	67 (91.8)	6 (8.2)	
Total comorbidities				
Mean (SD)	3.1 (2.0)	3.1 (2.0)	2.9 (1.8)	0.501
Hospital bed size				
Small	27 (6.9)	24 (88.9)	3 (11.1)	0.164
Medium	68 (17.3)	58 (85.3)	10 (14.7)	
Large	297 (75.8)	238 (80.1)	59 (19.9)	
Hospital teaching status				
Non-teaching	77 (19.6)	71 (92.2)	6 (7.8)	0.008
Teaching	315 (80.4)	249 (79.0)	66 (21.0)	
Outcomes (column percentages)				
Mortality	97 (24.7)	87 (89.7)	10 (10.3)	0.023
Poor outcome (54 observations missing)	166 (49.0)	138 (83.1)	28 (16.9)	0.175
Length of stay				
Mean (SD)	18.8 (18.1)	16.2 (16.1)	30.5 (21.7)	<0.0001
Total charges*				
Mean (SD)	199.7 (214.7)	170.8 (208.4)	327.1 (196.1)	<0.0001

*adjusted 3%/year for inflation in $1000.
DRG, Diagnosis Related Group; SAH, subarachnoid hemorrhage.

than in those who did not (p<0.0001; tables 1 and 2). Length of stay was 30.3 days in the clipping group and 20.8 days in the coiling group (p=0.074). Hospital charges were higher in the clipping group than in the coiling group (p=0.334).

DISCUSSION

IIAs are a rare entity and most of the clinical series provide the data from a small number of patients, limiting their usefulness for determining treatment-specific guidelines. To better understand current treatment practices for IIAs, we used the NIS database to provide the data from a large cohort of patients. This study of 393 patients with IIAs treated between 2002 and 2011 is the largest series of patients with IIA studied to date. Interestingly, most of the patients (82%) with ruptured IIAs were managed conservatively without surgical or endovascular treatment. This is surprising given that many practitioners feel obliged to treat ruptured IIA with embolization or clipping to prevent potentially fatal re-hemorrhage, particularly in the setting of systemic anticoagulation which may be necessary for this patient population. This study also demonstrates significantly higher mortality in the group of patients managed conservatively than in those who underwent aneurysm embolization or clipping. However, it cannot be determined whether this is secondary to a true benefit from intervention or rather more severe illness in those managed non-operatively. We attempted to control for illness severity by including APR-DRG, age, comorbidities, and other factors in a multivariate regression analysis; however, these metrics have limitations. APR-DRG has predominantly been used in non-surgical cohorts as a measure of overall illness severity,[10–13] limiting its relevance to this patient cohort. In addition, only 18% of patients underwent a corrective cardiac procedure. This study shows that in the USA, the vast majority of patients with IIA and IE are treated conservatively without aggressive neurosurgical or cardiothoracic interventions.

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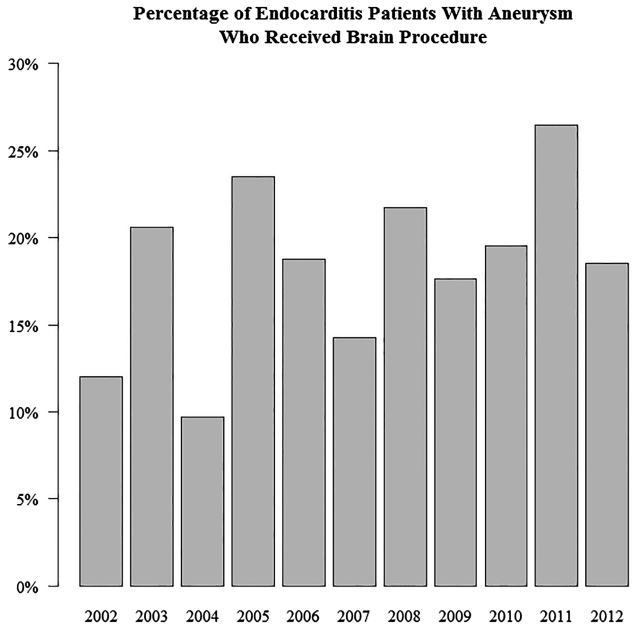

Figure 1 The percentage of patients with infectious intracranial aneurysms/infectious endocarditis receiving neurosurgical clipping or embolization for each study year.

The neurosurgical treatment of IIA is often challenging because these aneurysms are frequently small and located on small, distal arteries. Furthermore, the parent artery wall is often fragile, complicating reconstruction with surgical clips or endovascular techniques. Surgical options include parent artery occlusion with or without bypass and clipping of the aneurysm.[4 14 15] Earlier there was apprehension about the use of endovascular techniques such as coils and stents in the presence of infection;[16] however, these endovascular treatments have gained favor over the past few years, with one review showing no evidence suggesting continued infection due to the presence of coils and stents.[3] Endovascular strategies for IIAs include a direct approach by endovascular coiling with or without stent assistance,[17 18] embolization of the aneurysm with liquid embolic agents—for example, Onyx,[19 20] and parent artery occlusion with balloon/coils or liquid embolic agents, or a combination of these modalities.[21–25]

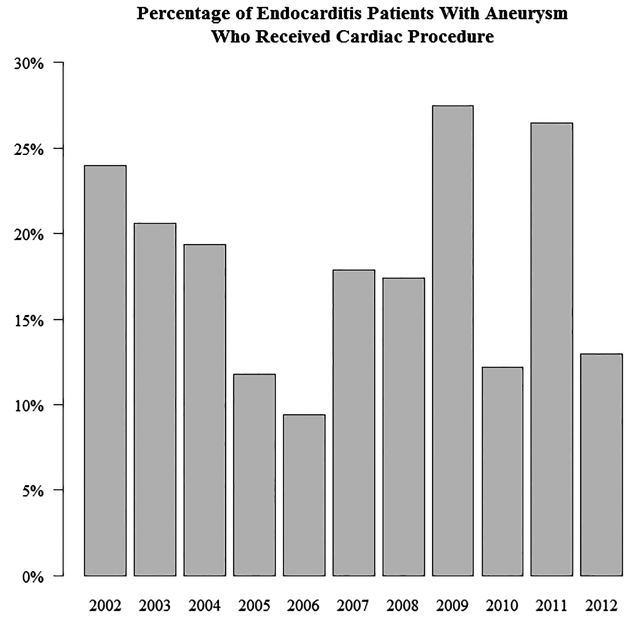

Figure 2 The percentage of patients with infectious intracranial aneurysms/infectious endocarditis receiving cardiac procedures per study year.

In this database study, most of the patients (63%) who underwent a neurosurgical intervention had endovascular embolization performed as the treatment. There was no significant difference in mortality and poor outcomes between the clipping and coiling groups in this database study. Larger studies are needed for outcomes comparison between coiling and clipping groups in the patients with IIAs. However, since many patients with IIAs have significant medical comorbidities, endovascular treatment may be the more feasible option. The need for anticoagulation and the higher risk of perioperative cardiac failure with surgical intervention may also make endovascular embolization the preferred neurosurgical treatment for IIAs.[26] One recent review article on the endovascular management of IIAs demonstrated high occlusion (95%) with low (5%) procedure-related complication rates.[27]

Recent reviews of the literature reveal the absence of widely accepted guidelines for the management of IIAs owing to the dearth of high-quality evidence.[3 27] The rupture rates for IIAs are believed to be higher than those of non-infectious cerebral aneurysms and range from 2% to 10%.[28 29] The published mortality rate for conservatively treated IIAs is as high as 80%.[30] Reports generally recommend immediate intervention for ruptured IIAs either by surgical or endovascular means, whereas unruptured IIAs may be treated with antibiotics alone with close follow-up.[3 27] Owing to the higher rate of rupture and the high mortality associated with their rupture, patients with certain unruptured aneurysms that are easily accessible and have a favorable morphology may be candidates for prophylactic neurosurgical treatment while receiving appropriate antibiotic therapy.[3 28] This analysis suggests that unruptured and ruptured aneurysms were treated with a similar frequency (25% vs 18%), though the number of unruptured aneurysms that was identified was much smaller (32 vs 361). Although these remain generally accepted treatment strategies for patients with an IIA, this study demonstrates a significant disparity between these suggested strategies and the way in which patients are being treated in the USA, particularly those with ruptured IIAs. The reasons for the very low percentage of patients in this NIS study undergoing neurosurgical aneurysm obliteration (19%) after having a SAH are unclear, but it is unlikely to be artifactual from procedural coding errors. Incorrect diagnosis coding might have led to a fraction of patients without a true IIA being included. It is perhaps more likely that most patients in the USA with ruptured IIAs are truly being managed conservatively. The reasons for this trend are unclear and require further investigation.

This study has several important limitations. NIS studies are retrospective, creating the potential for selection bias, and contain data from only 20% of all hospitalizations in the USA. NIS data are limited to only those variables recorded and is subject to coding errors or variability in coding. Owing to inherent limitations to the NIS, we are unable to compare outcomes based on the treatment types or understand the relationship between time course and treatments performed. Even with these limitations, the NIS provides unique insight into current treatment practices and outcomes in the management of IIAs unobtainable through other methods.

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CONCLUSIONS

In this NIS database study, the majority of patients with IIAs were managed non-operatively, regardless of rupture status. Patients undergoing neurosurgical procedures had better outcomes than those who did not during their hospitalizations, although the significance of this phenomenon is unclear. The proportion of the patients undergoing any intervention has remained constant over the years 2002–2011, although the number of patients with infective endocarditis and IIAs has increased. Further investigations seem warranted to standardize management to improve outcomes.

Contributors All the authors made significant contributions to the study and read and approved the final version of the manuscript.

Competing interests None declared.

Ethics approval Institutional review board, University of Florida.

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Amit Singla, Kyle Fargen, Spiros Blackburn, Dan Neal, Tomas D. Martin, Phillip J Hess, Thomas M Beaver, Charles T Klodell and Brian Hoh

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